AIoT FOR INTELLIGENT ENERGY

SmartGrid AIoT

Intelligent power monitoring & management system

A connected prototype combining an edge rover, a live grid twin, and AI-assisted operator response.



THE PROJECT

From physical sensing to a coordinated, explainable response.

EVENT

Power Hack 2026

Theme 5: AIoT for intelligent energy systems

PRESENTATION

18 November 2026

Presidency University + BESCOM Bengaluru

01 / PROJECT OVERVIEW

A grid story, made visible.

SmartGrid AIoT brings physical sensing, live grid visibility, and assisted operator response into one connected prototype.

01 / PHYSICAL EDGE

A mobile sensing rover

An ESP32-S3 rover patrols a miniature energy corridor, captures local conditions, and shares timestamped telemetry.

02 / DIGITAL TWIN

A live view of the grid

A web-based control room reflects the three modeled nodes, highlights changing states, and visualizes simulated power-flow changes.

03 / ASSISTED RESPONSE

A clearer operator decision

AI-generated summaries and voice alerts help explain a fault and prepare the next response for operator review.

THE CORE IDEA

Sense at the edge. Understand the network. Support a safe, human-approved response.

One connected story across hardware, software, and operator experience.

02 / SYSTEM ARCHITECTURE

One physical edge. One shared control room.

A simple, bidirectional path connects the rover to the digital twin

01
PHYSICAL EDGE
ESP32-S3 rover + modular sensors + local TFT status display.
SENSE

02
CLOUD + DATA
Wi-Fi telemetry flows through Next.js APIs into Supabase storage and Realtime.
STREAM

03
DIGITAL TWIN
React Flow models the three grid nodes and presents the operator control room.
VISUALIZE

04
AI + VOICE
Gemini via OpenRouter supports diagnostics and reports; ElevenLabs delivers spoken alerts.
EXPLAIN

RETURN PATH / OPERATOR TO EDGE

Operator-approved commands and live thresholds travel back through Supabase Realtime to the rover.

The system is designed as one telemetry-and-control loop.

03 / THE MINIATURE GRID

Three nodes. One shared grid story.

The same corridor anchors the physical patrol, digital twin, fault scenario, and operator narrative.

A

RESIDENTIAL FEEDER

Normal operating state

BASELINE

B

EV CHARGING HUB

Peak-load condition

PEAK LOAD

C

INDUSTRIAL TRANSFORMER

Thermal fault scenario

CRITICAL

ROVER

PATROL ROUTE

Node A -> Node B -> Node C

Sense locally, then publish node-linked telemetry.

ONE CONSISTENT MODEL

Node names and states stay aligned across the rover, landing story, control room, AI assistant, and workflow.

04 / CORE CAPABILITIES

From sensing to supported action.

Four capability groups turn the prototype into a coherent intelligent-energy demonstration.

01

EDGE SENSING + PATROL

A mobile rover gathers thermal, voltage/current, and event signals while navigating the tabletop corridor.

02

REAL-TIME GRID VISIBILITY

Telemetry updates node states in the digital twin, making normal, peak-load, and fault conditions easy to distinguish.

03

AI + VOICE ASSISTANCE

Recent sensor logs support diagnostic summaries and maintenance reporting; critical events can be announced aloud.

04

OPERATOR WORKFLOWS

Thresholds and response steps can be reviewed, tested, and published through a visual workflow experience.

HUMAN-IN-THE-LOOP

AI recommends and explains. The operator remains responsible for reviewing, testing, and publishing the response.

05 / END-TO-END DEMONSTRATION

One fault. A synchronized response.

The tabletop sequence makes the physical event and its digital consequences visible as one chain.

01

START PATROL

The operator starts the rover from the control room.

02

PATROL + SENSE

The rover moves from Node A through Node B toward Node C, streaming telemetry.

03

TRIGGER THE EVENT

A controlled thermal or load anomaly is introduced at Node C using a low-voltage demo input.

04

EDGE RESPONSE

The rover stops, its TFT switches to a critical alert, and a relay simulates isolation.

05

TWIN + AI RESPONSE

The twin marks Node C critical and animates a simulated reroute; AI summarizes the event and prepares a dispatch brief, with voice alert.

DEMO BOUNDARY

Breaker isolation and power rerouting are simulated on a safe, isolated low-voltage tabletop setup.

06 / OPERATOR EXPERIENCE

A calmer, clearer operator experience.

The product experience moves from public explanation to a focused workspace where people can understand, review, and coordinate a response.

01 / DISCOVER

Editorial landing story

A six-chapter visual introduction follows the rover from observation through response and into the control room.

02 / OPERATE

Three-node control room

A simplified digital twin exposes live node states and three primary actions: Start Patrol, Test Scenario, and Publish Changes.

03 / REASON

AI assistant + artifacts

Operators ask about the live grid, review a response recommendation, and keep reports, snapshots, and voice previews in context.

04 / ORCHESTRATE

Free-form workflow canvas

Response steps can be arranged, connected, tested, and published without losing the active session context.

VISUAL LANGUAGE

Warm paper. Deep ink. Live signal.

CONTROL HEALTHY WARNING CRITICAL

Large touch targets, clear labels, accessible interactions, and reduced-motion fallbacks.

07 / SUBMISSION SUMMARY

A focused prototype. A credible energy story.

SmartGrid AIoT demonstrates how edge sensing, a shared grid view, and operator-centred intelligence can work together in one connected experience.

VISIBILITY

Bring local conditions and changing node states into one live view.

COORDINATION

Connect a physical event to an understandable, synchronized response.

DECISION SUPPORT

Turn recent telemetry into concise summaries and reviewable next steps.

WHAT THE DEMONSTRATION MAKES VISIBLE

- A physical rover and modular sensing layer
- Live three-node digital twin and telemetry
- Fault highlighting with simulated rerouting
- AI maintenance summary and voice alert
- Operator review, test, and publish workflow

PROTOTYPE + SAFETY BOUNDARY

This is a prototype demonstration, not a live utility-grid control system. Use isolated, low-voltage demo loads only. Do not connect the rover, sensors, relay, or tabletop setup to mains wiring or a real distribution network.

Prepared from Team Celestia Playbook V3.0 and the approved experience update, 07 Nov 2026.

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